

Alias: Corbato

Homework 2 Functions

Unset

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Main:  add  $t0,$s1,$s1      # $t0 ← 2y
        add  $t0,$t0,$t0    # $t0 ← 4y
        addi $t0,$t0,4      # $t0 ← 4y+4
        addi $sp,$sp,-4     # Push $t0
        sw   $t0,0($sp)
        add  $a0,$t0,$zero   # arg0 ← 4y+4
        jal  Foo            # Call foo
        lw   $t0,0($sp)     # Pop $t0
        addi $sp,$sp,4      #
        add  $s0,$v0,$v0    # x ← 2*foo(4*y+4)
        add  $s0,$t0,$s0    # x ← x+4*y+4
        ...                # End of Main

Foo:    addi $sp,$sp,-104    # (MOD 1) Stack space for $ra, $fp,
                           # magic, junk
        sw   $ra,100($sp)   # Push $ra
        sw   $fp,96($sp)   # Push $fp
        addi $fp,$sp,100    # Establish frame
        addi $t1,$zero,0    # $t1 ← 0
        sw   $t1,-44($fp)   # junk[0] ← 0
        addi $t1,$zero,1    # $t1 ← 1
        sw   $t1,-40($fp)   # junk[1] ← 1
        ...
        addi $t1,$zero,9    # $t1 ← 9
        sw   $t1,-8($fp)    # junk[9] ← 9

        lw   $t1,-28($fp)   # $t1 ← junk[4]
        lw   $t2,-68($fp)   # (MOD 2) $t2 ← magic[8]
        sub  $t2,$zero,$t2  # $t2 ← -magic[8]
        addi $t2,$t2,2      # $t2 ← 2-magic[8]
        add  $t1,$t1,$t2    # $t1 ← junk[4]+2-magic[8]
        sw   $t1,-68($fp)   # junk[4] ← junk[4]+2-magic[8]
        slti $t0,$a0,1      # $t0 ← 1 if n<1; else 0
        bne  $t0,$zero,RetOne # Return 1 if n<1
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# (MOD 3) Call bar(n+1,magic[8],n+2,n+3,magic[8])
    addi $sp,$sp,-20      # Stack space for 5 args
    addi $t0,$a0,1        # $t0 ← n+1
    sw    $t0,0($sp)      # Push n+1
    lw    $t1,-68($fp)    # $t1 ← magic[8]
    sw    $t1,4($sp)      # Push magic[8]
    addi $t0,$t0,1        # $t0 ← n+2
    sw    $t0,8($sp)      # Push n+2
    addi $t0,$t0,1        # $t0 ← n+3
    sw    $t0,12($sp)     # Push n+3
    sw    $t1,16($sp)     # Push magic[8]
    jal   Bar              # Call bar(n+1,magic[8],n+2,n+3,magic[8])
    add   $t0,$v0,$v0     # $t0 ←
                          # 2*bar(n+1,magic[8],n+2,n+3,magic[8])
    addi $sp,$sp,20      # reclaim stack space

# Call foo(n-1)
    addi $sp,$sp,-4      # Push
    sw    $a0,0($sp)     # ... $a0
    addi $a0,$a0,-1      # arg0 ← n-1
    jal   Foo            # Call foo(n-1)
    lw    $a0,0($sp)     # Pop $a0
    addi $sp,$sp,4
    lw    $t1,-28($fp)   # $t1 ← junk[4]
    add   $v0,$v0,$t1    # $v0 ← foo(n-1)+junk[4]
    add   $v0,$v0,$t0    # (MOD 3) $v0 ← foo(n-1)+junk[4]+
                          # 2*bar(n+1,magic[8],n+2,n+3,magic[8])

    j     Ret            # Return
RetOne: addi $v0,$zero,1  # $v0 ← 1
Ret:     lw    $fp,96($sp) # (MOD 1) pop $fp
        lw    $ra,100($sp) # pop $ra
        addi $sp,$sp,104  # reclaim stack space
        jr    $ra        # return to caller

```